Machine Learning for Scientific Discovery: The Solvability Frontier

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Abstract

A first pass, the Solver Gap Engine of 2026-09-30, ranked 1,364 open problems from the formal-conjectures repository by their nearest solved problem in another file and found 31 at cosine similarity 0.8 or above [empirical: Solver Gap Engine stats.json at formal-conjectures 7d450ef, 2026-09-30, python3 match.py]. The frontier is the second pass. We embed 5,209 problem statements across seven canon branches with a small sentence encoder, call a problem's reach its highest similarity to a solved problem, and draw a frontier at the 10th percentile of solved problems' reach, 0.762 on this set [empirical: output/solvability-frontier/report/frontier.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`]. Open problems are sorted into a reach class: 268 close to known results, 863 borderline and 954 that need a new idea [empirical: output/solvability-frontier/report/predictions.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.t`
A backtest at cutoffs 2005 and 2021, and three frozen forecasts at 2005, 2015 and 2021, score the rule on the questions posed by those years under two outcome codings. Settled only (solved in 2026): at 2005 the inside rows settle at 24% against 5.9% outside (161 and 68 rows, ratio 4.012, permutation $p = 0.002$, AUC 0.799), at 2015 at 22% against 3% (222 and 66 rows, $p < 0.001$, AUC 0.812) and at 2021 at 13% against 1% (272 and 167 rows, ratio 21.489, $p < 0.001$, AUC 0.906). Of the scored rows, 11 at 2005 and 21 at 2021 are solved rows with no resolved year; without them the 2005 gap holds at 18% against 5.9% ($p = 0.020$, AUC 0.742) and the 2021 gap shrinks to 6% against 1% ($p = 0.013$, AUC 0.836), and placing the 90 undecided 2005 rows outside reverses the 2005 comparison to 24% against 27% [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`]. Settled or advanced (solved or partial) shows no gap: at 2005, 48% inside against 59% outside ($p = 0.153$, AUC 0.524), at 2015, 49% against 52% ($p = 0.777$, AUC 0.554), at 2021, 43% against 40% ($p = 0.615$, AUC 0.543) [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`]. The rule ranks settled problems above unsettled ones and does not rank partial progress, and partial is the uncertain column: the ingest assigned it in 2026 from a status page with no rule for how much progress counts. The embeddings and the 50 stored neighbours come from the 2026 corpus and the embedding model postdates every cutoff, so the frozen forecasts are a retrospective check under that leak; a prospective test needs embeddings fitted on pre-cutoff text and a neighbour set frozen per cutoff.

1 Introduction

Bucket's solvability atlas lists open and closed problems with their statements, branch, status and, where known, the years they were posed and resolved. The question behind the frontier is whether an open problem that sits close in meaning to solved problems is closer to being solved, and whether an AI system working from known results could reach it. The frontier drawing answers the first

clause by a rule; this paper tests the rule on problems whose outcome is known and publishes the ranked predictions for the open ones, with every number tagged to the file and command that reproduces it.

2 Origin: the Solver Gap Engine

The frontier began as a narrower tool. The Solver Gap Engine of 2026-09-30 read the google-deepmind formal-conjectures repository at commit 7d450ef, mapped each Lean declaration with its docstring and status into one row, and asked which open problems sit closest in wording to a solved one. It mapped 3,599 problems, 1,403 of them open [empirical: Solver Gap Engine stats.json at formal-conjectures 7d450ef, 2026-09-30, python3 match.py]. Rows whose docstring ran to 30 characters or fewer were dropped, leaving 1,364 open problems to rank [empirical: Solver Gap Engine stats.json at formal-conjectures 7d450ef, 2026-09-30, python3 match.py]. Each text was embedded with all-MiniLM-L6-v2 and each open row was scored by the cosine similarity of its nearest solved row in a different source file; a match in the same file was recorded and excluded, since two declarations in one file share a theme and a vocabulary by construction. The same-file nearest row was the closest overall for 455 of the 1,364 [empirical: Solver Gap Engine stats.json at formal-conjectures 7d450ef, 2026-09-30, python3 match.py].

The engine found 5 open problems at similarity 0.9 or above and 31 at 0.8 or above, with a median nearest cross-file similarity of 0.586 against a median of 0.167 for a random open-solved pair [empirical: Solver Gap Engine stats.json at formal-conjectures 7d450ef, 2026-09-30, python3 match.py]. The ranked list was meant as a queue for a prover: a high score marks a problem worth handing to one and says nothing about its difficulty. A Lean check script ran a sample of classical results against the mapped rows and exited non-zero on any failed check.

Three limits of that pass set the design of this one. It covered one repository and one branch, mathematics. The same-file exclusion was a rule of thumb for a leak it could name and no test of whether the score tracked outcomes. And it had no threshold: a ranked list with no line between inside and outside gives no count of what is reachable. The frontier keeps the embedding-and-nearest-solved core, widens the set to 5,209 rows across seven branches, replaces the file rule with a threshold fitted on the solved rows, and adds a backtest. The engine's code and its ranked table sit in a private intake folder pending release; the numbers above are its published summary, dated.

3 Related work

The atlas draws on the 2005 Science list of 125 questions [1, 2], the 2021 Science and Shanghai Jiao Tong University list[1], the google-deepmind formal-conjectures repository[2] and the Wikipedia lists of unsolved problems[3]. The encoder is bge-small-en-v1.5, whose model card points at the C-Pack paper [3]. Systems that generate and test scientific hypotheses end to end [4, 5] are the setting in which a reach score would be used; this paper measures only whether the score tracks outcomes.

[1] https://www.science.org/content/resource/125-questions-exploration-and-discovery, read from a Wayback copy of the booklet; no DOI.

[2] https://github.com/google-deepmind/formal-conjectures, Apache-2.0; no DOI.

[3] https://en.wikipedia.org/wiki/Lists_of_unsolved_problems, CC BY-SA 4.0; no DOI.

4 Preliminaries

Definition 4.1 (Set)**.** The set is the 71 atlas problems and the 5,138 rows of `problems-sourced.tsv`, 5,209 rows once Lean theorems are left off [empirical: output/solvability-frontier/report/frontier.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`]. A top-level row is solved when its status is solved. A variant is solved only when it and its parent are both solved. Each row's statement is embedded with bge-small-en-v1.5 at revision 5c38ec7, and its 50 most similar rows are stored with cosine similarity to three decimals [empirical: output/solvability-frontier/report/frontier.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Definition 4.2 (Reach)**.** The reach of a row is its highest similarity to a solved row other than itself, taken over its stored neighbours and its stored nearest solved row.

Definition 4.3 (Frontier)**.** The frontier threshold τ is the 10th percentile of reach over the solved rows. A row with reach at or above τ is inside; below it, outside. An open row whose branch holds fewer than 10 solved rows is unsampled and counted outside.

Definition 4.4 (Growth)**.** For a top-level row beyond the frontier, growth is one plus the number of beyond top-level rows among its 50 stored neighbours whose similarity to it is at or above τ: the number of rows that would move inside if it were solved. Variants neither pull nor count as pulled.

Definition 4.5 (Reach class)**.** Let u be the 75th percentile of reach among open rows inside the frontier. An open top-level row is "close to known results" when inside with reach at or above u, "borderline" when inside below u, "needs a new idea" when beyond, and "unsampled" when its branch holds fewer than 10 solved rows. Within a class, rows rank by growth, then reach, then id; a ranking by score with unique ids has one ordering [bm:BucketMath.Ranking.rank_ordered].

Definition 4.6 (Backtest)**.** For a cutoff year c, the solved set S_c holds every row solved under 4.1 with a resolved year at or before c. The tested set holds every row posed at or before c and outside S_c. Reach is computed as in 4.2 against S_c alone. When none of a row's stored neighbours is in S_c, its reach is unknown and bounded above by its 50th neighbour's similarity. The threshold τ_c is the 10th percentile of reach over the rows of S_c that have a neighbour in S_c; the rest are counted and dropped. A tested row is inside when its reach is at or above τ_c, outside when its reach or its bound is below τ_c, and undecided otherwise; undecided rows are counted and left out of the rates. The outcome is the row's 2026 status under two codings reported side by side: "settled" counts solved only, "settled or advanced" counts solved or partial. A solved row with no resolved year that was posed by the cutoff is tested and scored as resolved; every table repeats its first row with those rows removed. Undecided rows are also scored under two extremes, all inside and all outside, as bounds without a p-value.

5 Method

Statistics. Per cutoff and per stratum we report the inside and outside counts, the resolved rate on each side, their ratio (inside over outside, undefined when the outside rate is zero [bm:BucketMath.Marketing.ratio]) and a two-sided permutation p-value: the outcomes are shuffled 10,000 times under seed 20261006 and p is the share of shuffles whose absolute rate difference is at least the observed one, with one added to numerator and denominator [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`]. The p-value so defined is positive and at most one (C). A stratum with fewer than 10 rows on either side reports counts only. The AUC is the Mann-Whitney probability that a resolved row's reach exceeds an unresolved row's, ties counting one half.

Strata. Rows are stratified by the word count of the embedded text, under 20, 20 to 40 and over 40 words, and by branch. The bands partition the naturals (C). The length bands exist because the dated lists are short paraphrases: their statements average 19 words against 34 for the rest of the file [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Reproduction. From the repository root:

```
bash tools/solvability-atlas/build.sh
npx ts-node scripts/build-solvability-frontier-report.ts
python3 tools/solvability-atlas/makeup.py --publish
cd papers/solvability-frontier && make figures && make pdf
```

The first command rebuilds the embeddings and the neighbour file. The second writes `frontier.json`, `backtest.json`, `predictions.json` and `predictions.csv` under `output/solvability-frontier/report/` and the page data under `src/lib/research-os/`. The third writes the make-up counts. `make figures` redraws every figure and table from those files. The report files `backtest.json`, `predictions.csv`, `frontier.svg` and `makeup.json` are published beside this paper at https://www.bucket.foundation/research/papers/solvability-frontier.

6 Provenance and method

6.1 Sources

1 lists every source the rows came from, with its licence, what was read from it and what the ingest did to it. The 5209 rows are 71 curated atlas problems and 5138 ingested rows [empirical: output/solvability-frontier/report/makeup/makeup.json, 2026-10-08, python3 tools/solvability-atlas/makeup.py –publish]. 1 shows the make-up of the set under every label the pipeline reads: mathematics holds 4,112 rows, formal-conjectures supplies 3,567, Apache-2.0 covers 3,557, 687 rows carry a posed year, 309 a resolved year, and 2,194 are variants of another row [empirical: output/solvability-frontier/report/makeup/makeup.json, 2026-10-08, python3 tools/solvability-atlas/makeup.py –publish].

Table 1: Sources of the rows, with licence, what was taken and what was done [empirical: output/solvability-frontier/report/makeup/makeup.json, 2026-10-08, python3 tools/solvability-atlas/makeup.py –publish].

Source	Rows	Licence	What was taken	What was done to it
formal-conjectures	3567	Apache-2.0 (3557); Apache-2.0 and CC BY-SA 4.0 (10)	statement (text field, or the Lean declaration when empty), status, family and declaration suffix as name; 87 posed and 39 resolved years	deduped by normalised title; suffixed declarations marked as variants of their file's top-level row; keywords mined from the statement; years from the text only
Wikipedia unsolved lists	992	CC BY-SA 4.0	full list item as statement, bold or linked head as name, section heading as status; 37 posed and 56 resolved years	branch per page, market per list, deduped against the atlas and within the file; years from the text only
Hilbert's problems	23	CC BY-SA 4.0	explanation cell, table status and year columns, 1900 as posed year; 23 posed and 6 resolved years	partial for partial, no consensus, disputed, weaker form or partially cells; curator overrides on 14 and 18
Smale's problems	14	CC BY-SA 4.0	explanation cell, table status and year columns, 1998 as posed year; 14 posed and 0 resolved years	same table rule; curator overrides on 8, 14 and 17
Wikipedia, other articles	60	CC BY-SA 4.0	open problems from articles outside the unsolved lists, each row with a verified quote; 8 posed and 2 resolved years	added for the thin branches; status open; branch per article; deduped against the atlas and within the file
dated lists (Science 2005 and 2021, DARPA, XPRIZE, neuroscience 23, Holy Grails)	325	AAAS copyright, headline cited, statement paraphrased (239); CC BY-SA 4.0 (37); ACS copyright, title cited, statement paraphrased (26); Library of Congress record, chapter titles cited (23)	headline under 15 words as name, list year as posed year, a 2026 status check per row; 325 posed and 34 resolved years	statement paraphrased in 15 to 40 words; deduped by title and at 0.85 similarity; solved rows carry the discovery year
solved timelines, Wikipedia and Kavli	157	CC BY-SA 4.0	a question shown to predate its answer, the discovery or proof year, a posed-evidence quote under 12 words; 122 posed and 157 resolved years	statement rewritten as the question stood before resolution; discovery rows with no prior question held back; 159 pages verified
atlas, problems.tsv	71	MIT	name, branch, level, Lean status, posed and resolved years, markets, keywords, all curated; 71 posed and 15 resolved years	resolved year marks solved; embedding text is the problem record when one exists

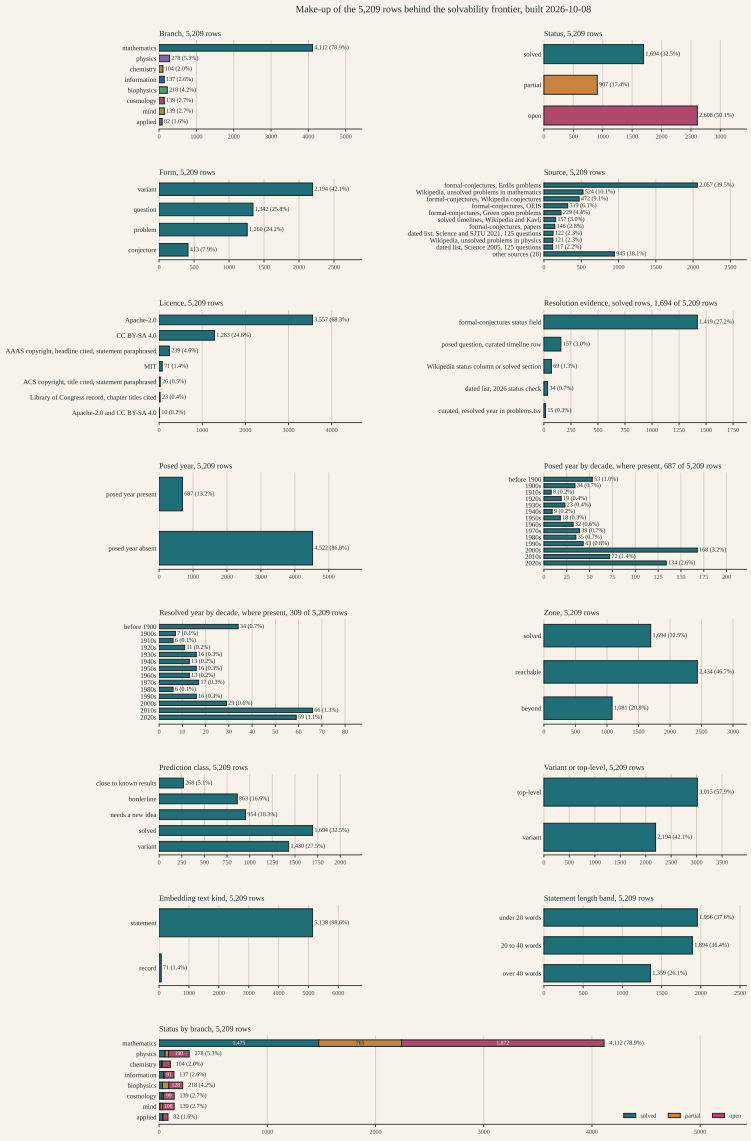


Figure 1: Make-up of the 5,209 rows by branch, status, form, source, licence, resolution evidence, posed and resolved years, zone, prediction class, variant or top-level, embedding text kind and statement length, with status by branch [empirical: output/solvability-frontier/report/makeup/makeup.json, 2026-10-08, python3 tools/solvability-atlas/makeup.py –publish].

Reading the figure. Every number in this box: [empirical: output/solvability-frontier/report/makeup/makeup.json, 2026-10-08, python3 tools/solvability-atlas/makeup.py –publish]. Mathematics holds 4,112 of the 5,209 rows, and the formal-conjectures repository supplies 3,567 of them, 2,057 from the Erdős problems alone, which is why Apache-2.0 covers 3,557 rows and the licence panel mirrors the source panel. Status is 1,694 solved, 907 partial and 2,608 open, and 2,194 rows are variants of another row, so the 1,430 open variants carry no class of their own. Only 687 rows carry a posed year and 309 a resolved year, with the posed years piling into the 2000s (168) and 2020s (134) because the two Science lists and the recent repository entries are the dated cohorts. The resolved-year evidence is the formal-conjectures status field for 1,419 rows and a curated, timeline or dated check for the remaining 275, and 3,458 rows have no resolution evidence at all. Statement length splits into 1,956 rows under 20 words, 1,894 between 20 and 40 and 1,359 over 40, the bands the backtest stratifies on.

6.2 Pipeline

Each step names its formula; i and j index rows.

1. **Text.** t_i is the ingested statement when present, else the atlas problem record (name, aliases, statement, keywords, key work titles), else name plus keywords: 5,138 statement, 71 record and 0 name plus keywords rows [empirical: output/solvability-frontier/report/makeup/makeup.json, 2026-10-08, python3 tools/solvability-atlas/makeup.py –publish]. The branch word never enters the text.

2. **Embedding.** $e_i = f(t_i)/\|f(t_i)\|$ with f = bge-small-en-v1.5 at revision 5c38ec7, 384 dimensions, cached by model revision and text hash [empirical: output/solvability-frontier/report/frontier.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

3. **Similarity.** $s_{ij} = e_i \cdot e_j$, the cosine. The 50 highest s_{ij} per row are stored to three decimals, with the nearest solved row over the whole set [empirical: output/solvability-frontier/report/frontier.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

4. **Solved set.** $S = \{\, i : \text{status}_i = \text{solved} \wedge (\text{parent}_i = \varnothing \vee \text{status}_{\text{parent}_i} = \text{solved}) \,\}$, so a proved special case of an open problem stays partial; $|S| = 1{,}690$ [empirical: output/solvability-frontier/report/frontier.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

5. **Reach.** $r_i = \max_{j \in S,\, j \neq i} s_{ij}$ over the stored neighbours and the stored nearest solved row (4.2).

6. **Threshold.** $\tau = Q_{0.1}(\{r_j : j \in S\})$, the value at index $\lfloor 0.1\,|S| \rfloor$ of the sorted solved reaches; $\tau = 0.762$ [empirical: output/solvability-frontier/report/frontier.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

7. **Zones.** solved if $i \in S$; unsampled if the branch holds fewer than 10 solved rows; reachable if $r_i \geq \tau$; beyond if $r_i < \tau$: 1,694, 0, 2,434 and 1,081 rows [empirical: output/solvability-frontier/report/frontier.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

8. **Radius.** With $\text{span}(x, a, b) = \min(1, \max(0, (x-a)/(b-a)))$: $\rho_i = 0.6\,(1 - \text{span}(r_i, \min_S r, 1))$ for solved rows, $0.6 + 0.4\,(1 - \text{span}(r_i, \tau, 1))$ for reachable rows, $1 + 0.5\,(1 - \text{span}(r_i, \text{floor}, \tau))$ for beyond rows and $1 + 0.5\,(1 - \text{span}(r_i, \text{floor}, 1))$ for unsampled rows, where $\text{floor} = \min(\tau, \min_i r_i)$ [empirical: output/solvability-frontier/report/frontier.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-r`

9. **Growth.** $g_i = 1 + |\{\, j \in N_i : \text{zone}_j = \text{beyond},\ j \text{ top-level},\ s_{ij} \geq \tau \,\}|$ over the 50 stored neighbours N_i (4.4).

10. **Angle.** $p_i = (e_i - \bar{e})\, V_2$ with V_2 the first two right singular vectors of the centred embeddings; θ_i is the rank of $\text{atan2}(p_{i2}, p_{i1})$ over the rows, scaled to $[0, 2\pi)$ [empirical: output/solvability-frontier/report/frontier.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

11. **Reach class.** $u = Q_{0.75}(\{r_i : i \text{ open, inside}\}) = 0.852$ [empirical: output/solvability-frontier/report/predictions.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`]; close when $r_i \geq u$, borderline when $\tau \leq r_i < u$, needs a new idea when $r_i < \tau$, unsampled by branch (4.5).

12. **Backtest statistic.** At cutoff c: $d_c = p_{\text{in}} - p_{\text{out}}$ with p the resolved share of scored rows on each side of τ_c; ratio $p_{\text{in}}/p_{\text{out}}$, undefined when $p_{\text{out}} = 0$ [bm:BucketMath.Marketing.ratio]; AUC $= \Pr(r_{\text{resolved}} > r_{\text{unresolved}})$ with ties at one half (4.6).

13. **Permutation test.** $p = (1 + |\{b : |d_b^*| \geq |d_c|\}|)/(B+1)$ with $B = 10{,}000$ shuffles of the outcomes across the inside and outside rows under seed 20261006 [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`], so $0 < p \leq 1$ (C).

OpenAlex activity. A title-and-abstract search on each sourced row's name or first clause, run against the OpenAlex works index with a key, stores the matching work count, counts per year for the last 30 years and the three most cited works per row in `tools/solvability-atlas/openalex-sourced.json`. The free daily budget is 1,000 requests, two per row. The first run took the file in its own order and stored 499 of the 5,138 rows before the budget ran out, 388 of them with at least one matching work, all 499 in mathematics (median 15 works) and none in the other seven branches [empirical: output/solvability-frontier/report/makeup/makeup.json, 2026-10-08, python3 tools/solvability-atlas/makeup.py –publish]; a second run the same day was refused for ten hours. Later runs take the remaining rows in an order that interleaves the branches. Until every branch holds at least 30 rows with data no comparison across branches is possible. The frontier score does not use these counts yet; 1 carries no activity axis and the distribution by branch sits in the page's make-up charts.

7 Results

7.1 Coordinates

Radius. With $r_i = \max_{j \in S,\, j \neq i} s_{ij}$ and $\mathrm{span}(x, a, b) = \min(1, \max(0, (x-a)/(b-a)))$, the radius is piecewise [empirical: output/solvability-frontier/report/frontier.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontie`

$$\rho_i = \begin{cases} 0.6\,(1 - \mathrm{span}(r_i, \min_S r, 1)) & i \in S, \\ 0.6 + 0.4\,(1 - \mathrm{span}(r_i, \tau, 1)) & \text{reachable}, \\ 1 + 0.5\,(1 - \mathrm{span}(r_i, \mathrm{floor}, \tau)) & \text{beyond or unsampled}, \end{cases} \qquad \mathrm{floor} = \min(\tau, \min_i r_i).$$

Within each band ρ is a monotone rescaling of $1 - r$, and the frontier sits at $\rho = 1$. The bands have different scales, so equal radial distances in different bands are not equal differences in reach: the solved band maps $[\min_S r, 1]$ onto 0.6 units, the reachable band maps $[\tau, 1]$ onto 0.4, and the outside band maps $[\mathrm{floor}, \tau]$ onto 0.5.

Angle. The centred embeddings are projected on their first two principal components, which carry 9.8 and 3.7 percent of the variance [empirical: ~/.cache/bucket-atlas vectors for 5,209 of the 5,209 rows, 2026-10-08, SVD of the centred embeddings in python3]; atan2 of the two coordinates gives a raw angle, the rows are sorted by it, and θ spaces them evenly on $[0, 2\pi)$ [empirical: output/solvability-frontier/report/frontier.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`]. So neighbours on the circle are neighbours in the two leading components, arc length counts rows and says nothing about how different they are, and the angle sees only those two directions. The angle depends on the embedding alone and the radius on the solved set alone, so a row keeps its angle whether it is solved, inside or outside; a sector being solved shows as rows moving inward.

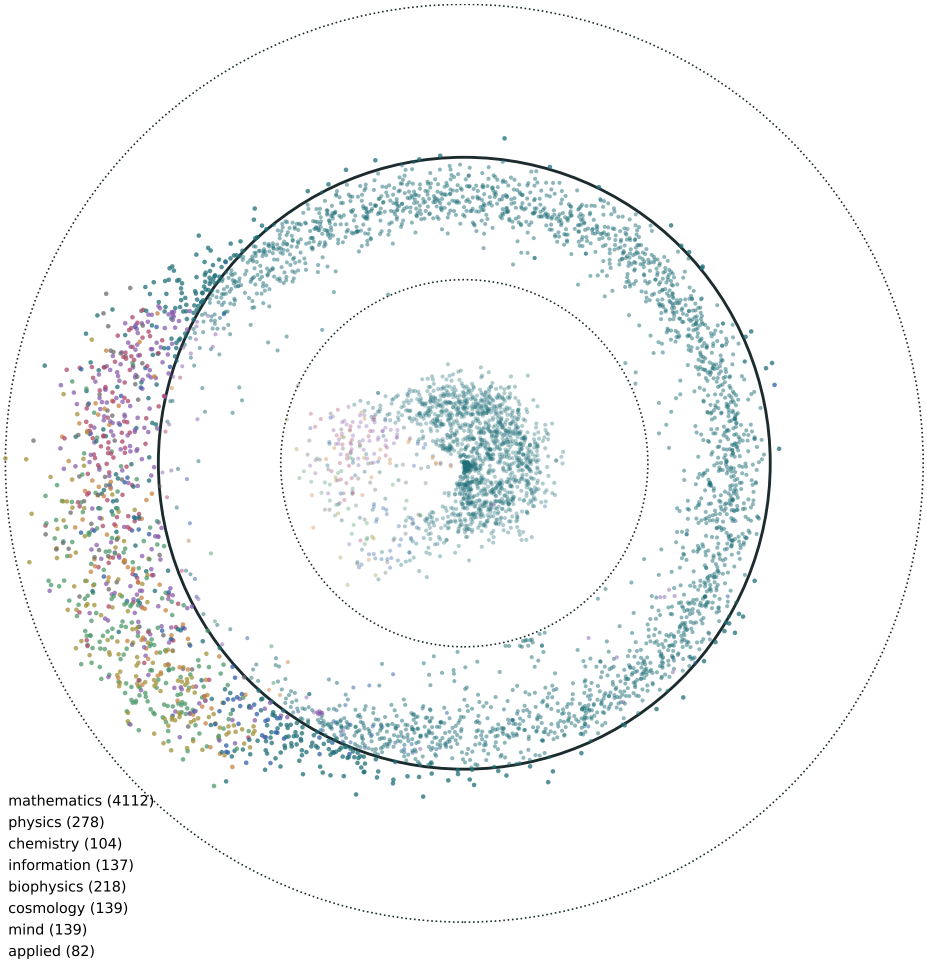


Figure 2: The frontier on 5,209 rows. Solved rows fill the inner disc by reach, open rows inside fill the ring up to the circle at 0.762, outside rows sit beyond it by how far their reach falls short, and any unsampled rows would be grey crosses [empirical: output/solvability-frontier/report/frontier.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Reading the figure. Every number in this box: [empirical: output/solvability-frontier/report/frontier.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].
A solved row's reach is its highest similarity to another solved row, and its radius in the inner disc is $0.6\,(1-\mathrm{span}(r, \min_S r, 1))$, so reach near 1 sits at the centre and the minimum solved reach, 0.57, at the dashed core ring, where solved results with no close solved sibling in the corpus sit. The exact centre holds solved rows with near-duplicate solved neighbours, proved variants of one Erdős problem and theorem families with near-identical statements: 182 solved rows have reach above 0.95 and 133 of them are variants. The centre carries no meaning of ease, importance or time, only density of solved work in statement space; and since the threshold is the 10th percentile of solved reach, many near-duplicate solved rows pull it up and tighten the circle. The circle holds 4,128 rows and 1,081 sit outside it: 1,694 solved, 2,434 reachable, 1,081 beyond and none unsampled. 3,825 of the inside rows are mathematics, which holds 1,475 of the 1,694 solved rows, so the solved base is a mathematics base; physics has 55 solved rows and 93 inside against 185 outside, biophysics 31 solved and 35 against 183, cosmology 28 solved and 32 against 107, and mind 10 solved and 10 against 129, a branch that held 5 solved rows, under the floor of ten, before the October additions. The ring is dense just inside the circle: 219 open rows have reach between 0.762 and 0.782, against 177 beyond rows between 0.742 and 0.762 and 638 below 0.70. The five outside top-level rows with the most growth are dynamical neuroscience (growth 14, reach 0.706, nearest solved: experience-driven reorganisation of the brain), computational neuroscience (14, 0.702, undirected connectivity in logarithmic space), the question of whether we can understand the action of the brain in natural environments (14, 0.695, chemical neurotransmission at synapses), the cosmological principle (13, 0.721, baryon acoustic oscillations) and connectionism (12, 0.706, undirected connectivity in logarithmic space). Of the atlas problems, Goldbach sits inside at reach 0.867, Riemann at 0.771 and abc just outside at 0.760, twin prime just outside at 0.745, and Navier-Stokes and dark matter further out at 0.671 and 0.678. The angle is a rank along the first two principal components, so rows cluster by vocabulary: 817 of the 830 rows in the first radian are mathematics, and 1,059 of the 1,097 rows outside mathematics sit between 2 and 4 radians, the left of the disc. Reach is text similarity, so a row inside shares words with a solved row and may share no method; and the embeddings and neighbours come from the 2026 corpus, so the drawing describes the corpus as it stands and forecasts nothing.

7.2 The frontier

The frontier sits at $\tau = 0.762$: 1,694 rows solved, 2,434 reachable and 1,081 beyond, 4,128 inside and 1,081 outside, with no branch unsampled now that mind holds 10 solved rows [empirical: output/solvability-frontier/report/frontier.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`]. 2 gives the split per branch. Mathematics holds 4,112 of the 5,209 rows and 1,475 of the 1,694 solved ones [empirical: output/solvability-frontier/report/frontier.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-re` so it sets the threshold on its own.

Table 2: Rows per branch and zone [empirical: output/solvability-frontier/report/frontier.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Branch	Total	Solved	Reachable	Beyond	Unsampled	Inside	Outside
mathematics	4112	1475	2350	287	0	3825	287
physics	278	55	38	185	0	93	185
biophysics	218	31	4	183	0	35	183
cosmology	139	28	4	107	0	32	107
mind	139	10	0	129	0	10	129
information	137	41	32	64	0	73	64
chemistry	104	22	5	77	0	27	77
applied	82	32	1	49	0	33	49

7.3 Backtest

3, 4, 5, 6 and 3 hold the backtest, and 8 the frozen forecasts at 2005, 2015 and 2021. At cutoff 2005, 170 rows were solved by then, 16 of them with no solved neighbour among the 50 stored and dropped from the threshold, which came out at 0.639; 319 rows were posed by 2005 and open at the cutoff, none unsampled, 90 undecided and 229 scored [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`]. At cutoff 2015, 213 rows were solved by then, 16 dropped, threshold 0.639; 351 rows tested, 63 undecided and 288 scored [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`]. At cutoff 2021, 259 rows were solved by then, 12 dropped, threshold 0.665; 482 rows tested, 43 undecided and 439 scored [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

The backtest is mixed. Under settled only the inside rows settle at a higher rate at all three cutoffs, and both the 2005 and the 2021 gaps survive removing the solved rows with no resolved year, the 2021 gap at a smaller size. Under settled or advanced the inside and outside rates are indistinguishable at every cutoff.

Settled only. At 2005 the inside rows settle at 24% over 161 against 5.9% over 68 outside (ratio 4.012, $p = 0.002$, AUC 0.799), and at 2021 at 13% over 272 against 1% over 167, 35 resolved against 1 (ratio 21.489, $p < 0.001$, AUC 0.906); 11 and 21 of those scored rows are solved today with no resolved year, and with them removed the inside rate is 18% against 5.9% (ratio 3.060, $p = 0.020$, AUC 0.742) at 2005 and 6% against 1% (ratio 9.315, $p = 0.013$, AUC 0.836) at 2021 [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`]. The settled rows number 42 at 2005 and 36 at 2021, and the undated solved rows in the tested set number 32 and 33 [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node`

scripts/build-solvability-frontier-report.ts]. At 2021 a little under half of the inside rate remains without those rows; at 2005 about three quarters of it does.

Settled or advanced. At 2005 the inside rate is 48% against 59% outside, ratio 0.813, $p = 0.153$, AUC 0.524; at 2021 it is 43% against 40%, ratio 1.063, $p = 0.615$, AUC 0.543 [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, npx ts-node scripts/build-solvability-frontier-report.ts]. With the undated solved rows removed: 44% against 59% ($p = 0.059$) and 38% against 40% ($p = 0.677$) [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, npx ts-node scripts/build-solvability-frontier-

Undecided rows. The 90 undecided rows at 2005 resolve at 42% settled and 61% settled or advanced; the 43 at 2021 at 30% and 56% [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, npx ts-node scripts/build-solvability-frontier-report.ts]. Placing all of them inside or all outside gives, settled only, inside rates of 30% or 24% against 5.9% or 27% at 2005 and 15% or 13% against 1% or 7% at 2021; settled or advanced, 53% or 48% against 59% or 60% at 2005 and 44% or 43% against 40% or 43% at 2021 [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, npx ts-node scripts/build-solvability-frontier-report.ts]. The last rows of each table carry the exact counts.

Table 3: Backtest at cutoff 2005, settled only, by stratum; the last two rows place the undecided rows all inside and all outside [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, npx ts-node scripts/build-solvability-frontier-report.ts].

Stratum	Inside	Outside	Resolved in	Resolved out	Ratio	p
all	161	68	24%	6%	4.012	0.002
all, undated solved rows removed	150	68	18%	6%	3.060	0.019
under 20 words	33	25	6%	4%	1.515	1.000
20 to 40 words	74	29	23%	3%	6.662	0.021
over 40 words	54	14	35%	14%	2.463	0.205
applied	8	3	under 10 on one side			
biophysics	29	33	10%	3%	3.414	0.331
chemistry	7	2	under 10 on one side			
cosmology	12	4	under 10 on one side			
information	20	2	under 10 on one side			
mathematics	51	1	under 10 on one side			
mind	6	12	under 10 on one side			
physics	28	11	32%	9%	3.536	0.234
all undecided inside	251	68	30%	6%	5.147	
all undecided outside	161	158	24%	27%	0.888	

Under settled or advanced two strata reach p below 0.05 and run against each other: at 2005 the 20 to 40 word band resolves at 54% inside against 76% outside ($p = 0.045$), while at 2021 the over 40 word band resolves at 47% inside against 0% outside over 27 rows ($p < 0.001$) [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, npx ts-node scripts/build-solvability-frontier-report.ts]. The second is a cohort effect: the long statements outside at 2021 are the 2021 list's own longer paraphrases, all still open five years on [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, npx ts-node scripts/build-solvability-frontier-report.ts]. Mathematics scores one row outside at 2005 and two at 2021 [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, npx ts-node scripts/build-solvability-frontier-report.ts], because nearly every dated mathematics row has a solved mathematics neighbour, so no mathematics rate is given.

Table 4: Backtest at cutoff 2005, settled or advanced, by stratum [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Stratum	Inside	Outside	Resolved in	Resolved out	Ratio	p
all	161	68	48%	59%	0.813	0.153
all, undated solved rows removed	150	68	44%	59%	0.748	0.059
under 20 words	33	25	48%	64%	0.758	0.288
20 to 40 words	74	29	54%	76%	0.713	0.045
over 40 words	54	14	39%	14%	2.722	0.120
applied	8	3	under 10 on one side			
biophysics	29	33	72%	76%	0.956	0.780
chemistry	7	2	under 10 on one side			
cosmology	12	4	under 10 on one side			
information	20	2	under 10 on one side			
mathematics	51	1	under 10 on one side			
mind	6	12	under 10 on one side			
physics	28	11	43%	27%	1.571	0.491
all undecided inside	251	68	53%	59%	0.894	
all undecided outside	161	158	48%	60%	0.795	

Table 5: Backtest at cutoff 2021, settled only, by stratum [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Stratum	Inside	Outside	Resolved in	Resolved out	Ratio	p
all	272	167	13%	1%	21.489	<0.001
all, undated solved rows removed	251	167	6%	1%	9.315	0.013
under 20 words	122	81	6%	0%		0.042
20 to 40 words	92	59	8%	2%	4.489	0.151
over 40 words	58	27	36%	0%		<0.001
applied	15	13	0%	0%		1.000
biophysics	41	50	2%	0%		0.451
chemistry	27	19	4%	0%		1.000
cosmology	19	11	0%	0%		1.000
information	23	8	under 10 on one side			
mathematics	96	2	under 10 on one side			
mind	24	34	0%	0%		1.000
physics	27	30	4%	0%		0.473
all undecided inside	315	167	15%	1%	25.448	
all undecided outside	272	210	13%	7%	1.930	

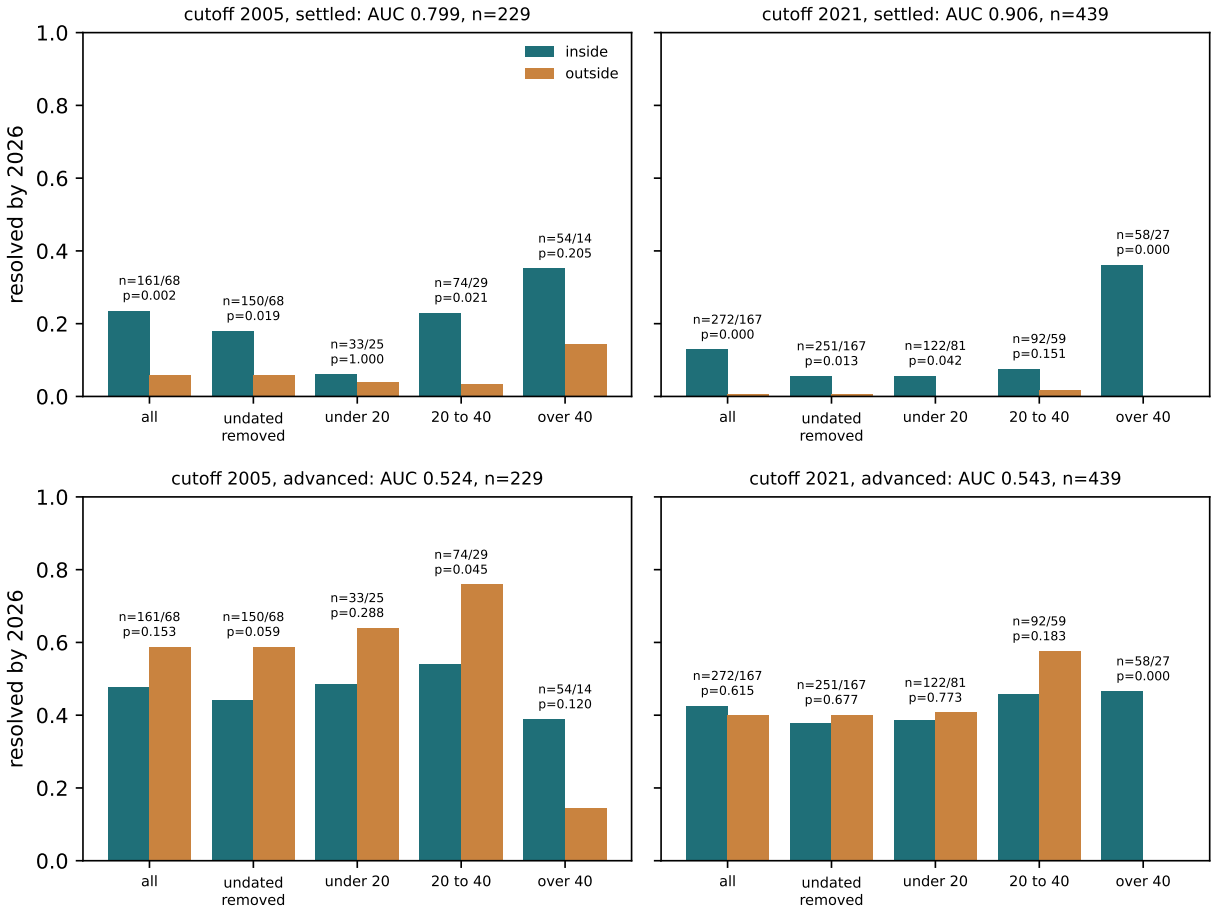


Figure 3: Resolved rate by 2026 inside and outside the cutoff frontier under both codings, over all scored rows, with the undated solved rows removed, and by length band [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Table 6: Backtest at cutoff 2021, settled or advanced, by stratum [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Stratum	Inside	Outside	Resolved in	Resolved out	Ratio	p
all	272	167	43%	40%	1.063	0.615
all, undated solved rows removed	251	167	38%	40%	0.943	0.677
under 20 words	122	81	38%	41%	0.946	0.773
20 to 40 words	92	59	46%	58%	0.792	0.183
over 40 words	58	27	47%	0%		<0.001
applied	15	13	13%	15%	0.867	1.000
biophysics	41	50	66%	66%	0.998	1.000
chemistry	27	19	18%	21%	0.880	1.000
cosmology	19	11	47%	27%	1.737	0.442
information	23	8	under 10 on one side			
mathematics	96	2	under 10 on one side			
mind	24	34	25%	44%	0.567	0.169
physics	27	30	26%	27%	0.972	1.000
all undecided inside	315	167	44%	40%	1.108	
all undecided outside	272	210	43%	43%	0.984	

7.4 October 2026 claims

Status on 2026-10-12. On 2026-10-06 OpenAI published 719 manuscripts in 372 result families, each a claimed result on an open problem, produced by an unreleased model [empirical: tools/solvability-atlas/claims/openai-math-2026-10.json, 2026-10-12, python3 tools/solvability-atlas/claims_build.py]. OpenAI's news feed stamps the announcement at 2026-10-06 12:00 UTC, a round figure that may be nominal, and the first commit of the repository at 21:58 UTC; the time of any earlier disclosure was not established [empirical: tools/solvability-atlas/claims/frontier-2026-10.json, 2026-10-12, python3 tools/solvability-atlas/claims_frontier.py]. The last atlas state committed before both dates from 04:07 UTC the same day, 7 hours 53 minutes before the feed timestamp, with 5,080 rows and a threshold of 0.781; the rows and neighbour file this paper reports were built and edited on 2026-10-07 and 2026-10-08, after the release [empirical: tools/solvability-atlas/claims/frontier-2026-10.json, 2026-10-12, python3 tools/solvability-atlas/claims_frontier.py]. The earlier state is therefore the one whose classes are a prediction made before the fact, dated against that feed timestamp, and 7 uses it, rebuilt from that commit with that commit's code [empirical: tools/solvability-atlas/claims/frontier-2026-10.json, 2026-10-12, python3 tools/solvability-atlas/claims_frontier.py]. The atlas records these rows under a status of its own, claimed, which every count in this paper reads as open: no named mathematician or refereed venue had confirmed a result of the batch by 2026-10-12, three manuscripts were withdrawn within a day, and the formalization catalogue marks its own review as unchecked [empirical: tools/solvability-atlas/claims/openai-math-2026-10.json, 2026-10-12, python3 tools/solvability-atlas/claims_build.py]. A hand mapping of the catalogue to the atlas finds 79 rows whose problem a family claims in full, 66 of them top-level rows of mathematics and information, the two branches the batch covers [empirical: tools/solvability-atlas/claims/frontier-2026-10.json, 2026-10-12, python3 tools/solvability-atlas/claims_frontier.py].

In the state before the release, inside rows are claimed at 5.2% and outside rows at 4.2%, a difference of +1.0 points with a 95% interval of −1.6 to 3.4, and reach ranks claimed rows above unclaimed ones with AUC 0.495, interval 0.426 to 0.574 [empirical: tools/solvability-atlas/claims/frontier-2026-10.json, 2026-10-12, python3 tools/solvability-atlas/claims_frontier.py]. Restricted to the 45 rows whose Lean statement covers the full problem, the class the atlas ranks first has the lowest rate, 2.4%

against 3.6% and 3.6%, with overlapping intervals [empirical: tools/solvability-atlas/claims/frontier-2026-10.json, 2026-10-12, python3 tools/solvability-atlas/claims_frontier.py]. The state this paper reports, frozen at the parent of the mapping six days after the release, with no commit or row that mentions the batch, gives the same reading as a comparison made after the fact: 4.9% inside against 5.0% outside, difference −0.1 points, interval −3.3 to 2.9, AUC 0.496 [empirical: tools/solvability-atlas/claims/frontier-2026-10.json, 2026-10-12, python3 tools/solvability-atlas/claims_frontier.py]. The two tests measure different outcomes on different cohorts: the 2021 backtest counts dated rows that the source lists record as solved over five years, 13% inside against 1% outside, and this table counts rows one lab claimed in one release. On this batch the inside and outside rates are level.

Table 7: Rows claimed settled in the OpenAI release of 2026-10-06, by reach class in the atlas state committed 7 hours 53 minutes before the announcement's feed timestamp. Open top-level rows of mathematics and information; 95% intervals from 2,000 bootstrap draws over groups of related rows [empirical: tools/solvability-atlas/claims/frontier-2026-10.json, 2026-10-12, python3 tools/solvability-atlas/claims_frontier.py].

		every full claim		Lean covers the full problem	
Class before the release	Open rows	Claimed	Rate, interval	Claimed	Rate, interval
close to known results	251	12	4.8%, 2.4 to 7.6	6	2.4%, 0.8 to 4.3
borderline	752	40	5.3%, 3.6 to 7.1	27	3.6%, 2.1 to 5.1
needs a new idea	337	14	4.2%, 2.0 to 6.5	12	3.6%, 1.7 to 5.6
inside the frontier	1,003	52	5.2%, 3.8 to 6.7	33	3.3%, 2.1 to 4.6
outside the frontier	337	14	4.2%, 2.0 to 6.5	12	3.6%, 1.7 to 5.6

Four limits bound the reading. The claims are unconfirmed, and a row leaves this table when its claim is withdrawn. One system attempted about 4,000 problems that its lab chose and has not listed, so the rows it claimed are a selected sample of what is solvable and the denominator here is the atlas, which the lab never saw. Ten of the 79 claimed rows carry a Lean file that covers less than the row, and 20 carry none [empirical: tools/solvability-atlas/claims/openai-math-2026-10.json, 2026-10-12, python3 tools/solvability-atlas/claims_build.py]. The mapping was made by hand from the catalogue text, after the release, and misses problems the atlas lists under another name. Among the 339 Erdős rows inside the frontier before the release the top class is claimed at 6.1% against 2.4% for borderline, with 19 outside rows and one claimed, which is too few to read [empirical: tools/solvability-atlas/claims/frontier-2026-10.json, 2026-10-12, python3 tools/solvability-atlas/claims_frontier.py]. The reading changes if confirmed claims, once at least 30 are recorded, settle inside at a rate whose interval for the difference from outside excludes zero, or if the reach AUC interval on confirmed rows clears 0.5; `tools/solvability-atlas/CLAIMS-2026-10.md` holds both baselines and the rule.

7.5 Frozen forecasts

8 holds one forecast per cutoff, written once to `tools/solvability-atlas/forecasts/` and scored afterwards. Each file lists the rows posed at or before its cutoff, the rows solved and resolved at or before it, its own threshold computed from that solved set alone, and an input hash over the rows it admitted. A row posed or resolved after the cutoff never enters, and the scorer reads the forecast and the current status column only. The 2005 and 2021 forecasts reproduce the live backtest above to the row, since the live computation applied the same filters; the 2015 forecast adds a third cutoff, with 213 rows solved by then, 351 tested and 288 scored [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

All three were built in 2026 with the outcomes already known: freezing fixes the method and the inputs, and this is a retrospective check rather than a prospective test. The embedding model postdates every cutoff. Settled rates count solved rows without a resolution year as resolved; the main backtest tables above carry the variant with those rows removed. At 2015 that variant gives 17% inside against 3% outside ($p = 0.006$, AUC 0.762) under settled only and 46% against 52% ($p = 0.475$, AUC 0.514) under settled or advanced [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Table 8: Frozen forecasts at three cutoffs, both codings, all scored rows [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Cutoff	Coding	τ_c	Inside	Outside	Resolved in	Resolved out	Ratio	p	AUC
2005	settled	0.639	161	68	24%	6%	4.012	0.002	0.799
2005	advanced	0.639	161	68	48%	59%	0.813	0.153	0.524
2015	settled	0.639	222	66	22%	3%	7.284	<0.001	0.812
2015	advanced	0.639	222	66	49%	52%	0.953	0.777	0.554
2021	settled	0.665	272	167	13%	1%	21.489	<0.001	0.906
2021	advanced	0.665	272	167	43%	40%	1.063	0.615	0.543

7.6 Where this is weak

Partial. The settled-only coding separates inside from outside at every cutoff and the settled-or-advanced coding separates nothing. Partial was assigned in 2026 by the ingest from a status page, with no threshold for how much progress counts, so the column that decides the result is the least reliable one in the file.

Leakage. The embeddings and the 50 stored neighbours come from the 2026 corpus. Statements written after a cutoff, including the 2021 list when scoring 2005, shape the geometry each cutoff is scored on, and the nearest solved row at a cutoff was found among neighbours chosen in 2026. The test checks the rule against known outcomes under that leak, which is a weaker claim than a forecast. The frozen forecasts fix the solved set, threshold and inputs per cutoff but share the 2026 embedding model, which postdates every cutoff. A clean version needs embeddings fitted on text dated at or before the cutoff and a neighbour set built from that text.

Vocabulary. Cosine similarity between statement embeddings measures shared words and phrasing. Two problems can share a method and no vocabulary, or vocabulary and no method.

Mathematics. 4,112 of 5,209 rows and 1,475 of 1,694 solved rows are mathematics [empirical: output/solvability-frontier/report/frontier.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`]. The threshold is a mathematics quantity, and the other six branches are read against it.

Two cohorts. Of the 687 rows with a posed year, 71 are atlas problems and 616 are sourced rows [empirical: output/solvability-frontier/report/makeup/makeup.json, 2026-10-08, python3 tools/solvability-atlas/makeup.py –publish]; 239 of the sourced rows come from the 2005 and 2021 Science lists, and those two lists supply most of the tested rows at both cutoffs [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`]. A list's paraphrases share length and

style, which is what the length bands are for, and the over 40 word result at 2021 shows the cohort rather than the rule [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

The 50-neighbour bound. At a cutoff with few solved rows, most of a row's solved peers fall outside its 50 stored neighbours, so its reach is unknown: 72 rows undecided and 16 solved rows dropped at 2005, 42 and 12 at 2021 [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`]. Dropping them can only raise the threshold; the undecided bounds above show how far they can move the inside rate.

Resolved years. Only 297 of the 1,694 solved rows carry a resolved year, so the solved set at a cutoff, 170 at 2005 and 259 at 2021, is a fraction of what was solved by then [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`], and 32 and 33 solved rows with no year sit in the tested set instead.

7.7 Predictions

9 gives the class counts: 268 rows close to known results, 863 borderline and 954 that need a new idea, with no row unsampled and $u = 0.852$ [empirical: output/solvability-frontier/report/predictions.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`]. 10, 11 and 12 list the first ten of each class by growth; the full ranked table with the three nearest solved problems and the starting works is `predictions.csv`. B lists the 56 open atlas problems in full. Among them, Goldbach and the Langlands program are close to known results, with reach 0.867 and 0.863 to an Erdős problem and to the geometric Langlands conjecture [empirical: output/solvability-frontier/report/predictions.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`]; Navier-Stokes, quantum gravity, the Hubble tension and dark matter head the "needs a new idea" class by growth, 5, 5, 4 and 4, with the aging problem at 3 [empirical: output/solvability-frontier/report/predictions.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`]. The Goldbach placement is the vocabulary effect of 7.6: the nearest solved row shares the words, and nothing in the method says whether it shares a method.

Table 9: Open top-level problems per reach class [empirical: output/solvability-frontier/report/predictions.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Class	Open problems
close to known results	268
borderline	863
needs a new idea	954
unsampled	0

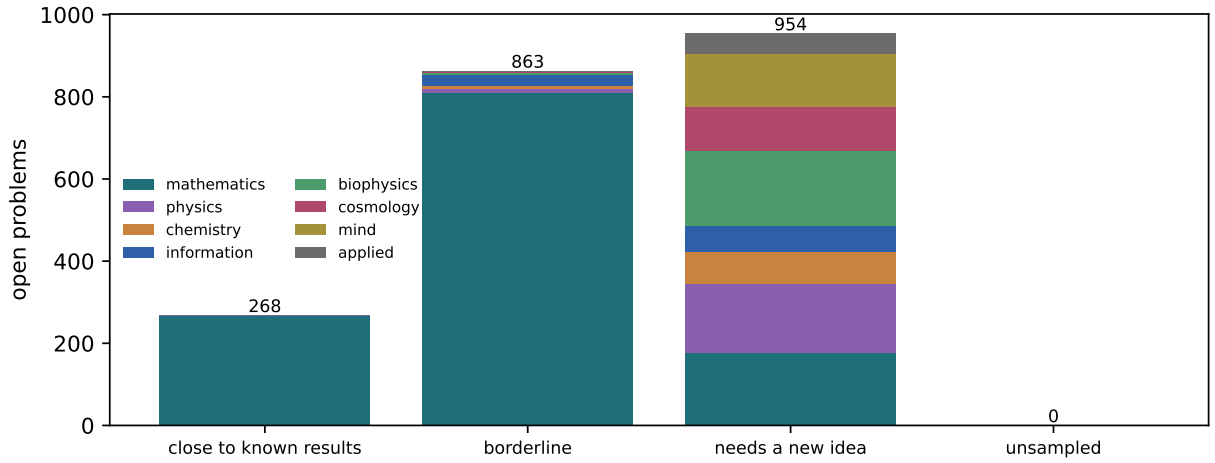


Figure 4: Reach classes by branch [empirical: output/solvability-frontier/report/predictions.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Reading the figure. Every number in this box: [empirical: output/solvability-frontier/report/predictions.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].
Mathematics holds 264 of the 268 rows close to known results and 811 of the 863 borderline rows, so the two inside classes are mathematics classes. Every other branch is mostly beyond: physics 168 of 178 top-level open rows, biophysics 183 of 187, chemistry 77 of 82, cosmology 107 of 111, applied 49 of 50 and mind 129 of 129 need a new idea under this rule, and information is the one branch outside mathematics with rows in the top class, 4 close and 28 borderline against 64 beyond. Mind has exactly the ten solved rows the floor asks for and none of its open rows reaches the frontier. The split follows the solved base: a branch with few solved rows has few neighbours to reach, so its open rows fall outside whatever their difficulty, and the class counts outside mathematics measure the corpus before they measure the problems.

Table 10: Close to known results, first ten by growth [empirical: output/solvability-frontier/report/predictions.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Problem	Branch	Reach	Growth	Nearest solved
OEIS A945, every prime occurs	mathematics	0.868	1	OEIS A363102, a eq one or prime
Goldbach conjecture	mathematics	0.867	1	Erdos problem 418, conditional
Erdos problem 881	mathematics	0.864	1	Erdos problem 35, plunnecke
Erdos problem 906	mathematics	0.861	1	Erdos problem 229
Ringel Conjecture	mathematics	0.977	0	Ringel Conjecture, large
Kotzig Conjecture	mathematics	0.970	0	Ringel Conjecture, kotzig large
Written on the Wall II problem GraphConjecture19	mathematics	0.960	0	Written on the Wall II problem GraphConjecture16
Written on the Wall II problem GraphConjecture40	mathematics	0.960	0	Written on the Wall II problem GraphConjecture58
OEIS A157237	mathematics	0.959	0	OEIS A157225
OEIS A38107	mathematics	0.958	0	OEIS A108, catalanReciprocalSum fracPart inj

8 Calibration

The central claims are empirical, so the calibration is the backtest itself: the rule is fixed before the outcomes are read, the outcomes are the 2026 status column with its per-row source, and the threshold and the strata are recomputed at each cutoff from rows dated at or before it. The error budget is the permutation p-value, the stratum floor, the two codings and the two bounded exclusions. Under settled only the inside rows settle at a higher rate at all three cutoffs, both gaps hold with the undated solved rows removed; under settled or advanced the AUC sits at 0.524, 0.554 and 0.543, each within 0.06 of chance [empirical: output/solvability-frontier/report/backtest.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`]. The reading is that the result turns on the partial column, and no prospective claim survives the embedding leak of 7.6.

9 Limitations

7.6 lists the weak points of the backtest and the frontier: the partial column, the 2026 embedding leak, vocabulary as similarity, mathematics dominance, the two cohorts, the 50-neighbour bound and the missing resolved years.

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References

[1] Donald Kennedy and Colin Norman. "What Don't We Know?" In: *Science* 309.5731 (2005), p. 75. DOI: 10.1126/science.309.5731.75.

Table 11: Borderline, first ten by growth [empirical: output/solvability-frontier/report/predictions.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Problem	Branch	Reach	Growth	Nearest solved
Cerny conjecture	information	0.776	4	Undirected connectivity in logarithmic space
Can all-pairs shortest paths be computed in strongly sub-cubic time, that is, in time for	information	0.766	4	Lovasz Plummer Conjecture, explicit
The Kobon triangle problem on triangles in line arrangements	mathematics	0.805	3	Erdos problem 619, h three le
Lebesgue's universal covering problem on the minimum-area convex shape in the plane that c	mathematics	0.789	3	Talagrand's convexity conjecture (Dongming Hua, Antoine Song, Stefan T
Are the nim-sequences of all finite octal games eventually periodic?	mathematics	0.788	3	Erdos problem 198, concrete
Finite lattice representation problem	mathematics	0.782	3	Modularity Conjecture
Banach Mazur Rotation	mathematics	0.772	3	Mazur's conjecture B (Vessilin Dimitrov, Ziyang Gao, and Philipp Habeg
Proton decay and spin crisis	physics	0.766	3	Internal structure of the proton
Dubner's conjecture	mathematics	0.841	2	Erdos problem 418, conditional
The Hadwiger?Nelson problem on the chromatic number of unit distance graphs	mathematics	0.833	2	Disproof of Hedetniemi's conjecture on the chromatic number of tensor

Table 12: Needs a new idea, first ten by growth [empirical: output/solvability-frontier/report/predictions.json, 2026-10-08, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Problem	Branch	Reach	Growth	Nearest solved
Dynamical neuroscience	mind	0.706	14	Experience-driven reorganisation of the brain
Computational neuroscience	biophysics	0.702	14	Undirected connectivity in logarithmic space
Can we understand the action of brain in natural environments?	mind	0.695	14	Chemical neurotransmission at synapses
Cosmological principle	physics	0.721	13	Baryon acoustic oscillations
Connectionism	mind	0.706	12	Undirected connectivity in logarithmic space
Free will, particularly the neuroscience of free will	mind	0.696	11	Keller's conjecture (Joshua Brakensiek, Marijn Heule, John Mackey, Dav
To what extent is the brain reconfigurable?	mind	0.686	11	Experience-driven reorganisation of the brain
Computational theory of mind	biophysics	0.677	10	Erdos problem 152
Consciousness	biophysics	0.661	9	Chemical neurotransmission at synapses
Cognition and decisions	biophysics	0.660	9	Experience-driven reorganisation of the brain

[2] Science News Staff. “So Much More to Know”. In: *Science* 309.5731 (2005), pp. 78–102. DOI: 10.1126/science.309.5731.78b.

[3] Shitao Xiao et al. *C-Pack: Packed Resources For General Chinese Embeddings*. The bge-small-en-v1.5 model card points at this paper. 2023. arXiv: 2309.07597 [cs.CL].

[4] Juraj Gottweis et al. *Accelerating scientific discovery with an AI co-scientist*. 2025. arXiv: 2502.18864.

[5] Chris Lu et al. “The AI Scientist: towards fully automated open-ended scientific discovery”. In: (2024). arXiv: 2408.06292.

A Glossary

reach https://github.com/gianyrox/bucket-foundation/blob/dev/src/lib/research-os/solvability-frontier.ts
A problem’s highest cosine similarity to a solved problem other than itself, over its stored neighbours; see 4.2.

frontier https://github.com/gianyrox/bucket-foundation/blob/dev/src/lib/research-os/solvability-frontier.ts
The reach value at the 10th percentile of solved problems’ reach; a problem at or above it is inside, below it outside; see 4.3.

growth https://github.com/gianyrox/bucket-foundation/blob/dev/src/lib/research-os/solvability-frontier.ts
For a problem beyond the frontier, one plus the number of beyond problems among its stored neighbours that sit at or above the threshold from it; see 4.4.

reach class https://github.com/gianyrox/bucket-foundation/blob/dev/src/lib/research-os/solvability-predictions.ts
One of four labels for an open problem: close to known results, borderline, needs a new idea, unsampled; see 4.5.

backtest https://github.com/gianyrox/bucket-foundation/blob/dev/src/lib/research-os/solvability-backtest.ts
The frontier rule applied at a past cutoff year to the problems posed by then, scored against their 2026 status; see 4.6.

B The atlas problems

Problem	Class	Branch	Reach	Growth	Nearest solved
Goldbach conjecture	reach	mathematics	0.867	1	Erdos problem 418, conditional
Langlands program	reach	mathematics	0.863	0	Geometric Langlands conjecture
Graph isomorphism complexity	borderline	information	0.833	0	Graph isomorphism in quasi-polynomial time
Jacobian conjecture	borderline	mathematics	0.807	0	Poisson Conjecture, jacobian implication

Problem	Class	Branch	Reach	Growth	Nearest solved
Collatz conjecture	borderline	mathematics	0.786	0	OEIS A306424, forty three is max
Riemann hypothesis	borderline	mathematics	0.771	0	Deterministic polynomial-time primality testing
Birch and Swinnerton-Dyer conjecture	borderline	mathematics	0.764	0	Elliptic Curve Rank, goldfeld two le
Navier-Stokes existence and smoothness	needs a new idea	physics	0.671	5	Poincare, smooth known cases
Quantum gravity	needs a new idea	cosmology	0.615	5	Microscopic Weighting, iff finite concentration
Hubble tension	needs a new idea	cosmology	0.707	4	Expansion of the universe
Dark matter identity	needs a new idea	cosmology	0.678	4	Baryon acoustic oscillations
BQP vs BPP separation	needs a new idea	information	0.693	3	Graph isomorphism in quasi-polynomial time
Black hole information paradox	needs a new idea	cosmology	0.691	3	Ultraviolet catastrophe
Origin of life	needs a new idea	biophysics	0.686	3	Urea synthesis from inorganic reagents
Weak cosmic censorship	needs a new idea	cosmology	0.659	3	Baryon acoustic oscillations
Hard problem of consciousness	needs a new idea	mind	0.630	3	Halting problem
Programmed versus stochastic aging	needs a new idea	biophysics	0.603	3	Urea synthesis from inorganic reagents
Room-temperature superconductivity mechanism	needs a new idea	physics	0.603	3	Element predicted as eka-aluminium
Glass transition	needs a new idea	chemistry	0.601	3	Continuum hypothesis independence
Hadwiger-Nelson problem	needs a new idea	mathematics	0.743	2	Erdos problem 780
Invariant subspace problem	needs a new idea	mathematics	0.727	2	Green open problem 49
Matrix multiplication exponent omega	needs a new idea	information	0.725	2	OEIS A108306
P vs NP	needs a new idea	information	0.722	2	Complexity of computing Nash equilibria
Protein folding pathway kinetics	needs a new idea	biophysics	0.721	2	Protein structure prediction
Elliptic curve discrete log hardness	needs a new idea	information	0.715	2	Modularity Conjecture
Existence of one-way functions	needs a new idea	information	0.711	2	Erdos problem 198, concrete
Yang-Mills mass gap	needs a new idea	physics	0.693	2	Microscopic Weighting, iff finite concentration
Cosmological constant problem	needs a new idea	cosmology	0.668	2	Erdos problem 1005, constant
Neural binding problem	needs a new idea	mind	0.661	2	Experience-driven reorganisation of the brain
AI alignment verification	needs a new idea	information	0.651	2	IBM Watson AI XPRIZE
Quantum measurement problem	needs a new idea	physics	0.648	2	Bose-Einstein condensate in a dilute gas
Turbulence closure	needs a new idea	physics	0.617	2	Voronovskaja Type Formula, bezier bernstein operators
Rational catalyst design	needs a new idea	chemistry	0.604	2	Sum Of Three Cubes, isSumOfThreeCubesRat any
Light, circadian biology and metabolism	needs a new idea	biophysics	0.585	2	Carrier of hereditary information
abc conjecture	needs a new idea	mathematics	0.760	1	Erdos problem 494, k eq 3 card gt 6
Integer factoring complexity	needs a new idea	information	0.752	1	OEIS A382590, kthPrimeFactor periodic
Unique Games Conjecture	needs a new idea	information	0.751	1	Complexity of computing Nash equilibria
Twin prime conjecture	needs a new idea	mathematics	0.745	1	OEIS A91669, prime and primitive root of dvd
Succinct proofs without trusted setup optimality	needs a new idea	information	0.722	1	Green open problem 14, W 3 20 lower

Problem	Class	Branch	Reach	Growth	Nearest solved
Hodge conjecture	needs a new idea	mathematics	0.714	1	Poincare conjecture
Hadamard matrix conjecture	needs a new idea	mathematics	0.706	1	Gerstenhaber Three Matrices, exists finrank adjoin quadruple
Exact exchange-correlation functional	needs a new idea	chemistry	0.695	1	Erdos problem 298, natural density
Bioelectric pattern control	needs a new idea	biophysics	0.677	1	Chemical neurotransmission at synapses
Autoformalization of mathematics	needs a new idea	information	0.667	1	Problem10 32, 10 32 strong
Fermion sign problem	needs a new idea	physics	0.666	1	Monochromatic Quantum Graph, eqSystem6 no solution d5
Formal reasoning limits of transformers	needs a new idea	information	0.659	1	Training networks over 1000 layers deep
Anomalous properties of water	needs a new idea	chemistry	0.659	1	Proof of the atomic theory
Smale's 18th problem limits of intelligence	needs a new idea	mind	0.656	1	Hilbert's tenth problem
Optimal multi-item auction design	needs a new idea	information	0.652	1	Complexity of computing Nash equilibria
Free energy principle formalization	needs a new idea	mind	0.649	1	Erdos problem 298, natural density
Market efficiency as computational hardness	needs a new idea	information	0.649	1	Erdos problem 298, natural density
Climate sensitivity bound	needs a new idea	physics	0.645	1	Sensitivity conjecture
Mitochondrial redox and aging mechanism	needs a new idea	biophysics	0.645	1	Carrier of hereditary information
2D Hubbard model ground state	needs a new idea	physics	0.640	1	Existence of time crystals (2012?2016)
Rough volatility pricing	needs a new idea	mathematics	0.626	1	Pricing of options
Plasma confinement stability	needs a new idea	physics	0.591	1	Bloch, landauConstant lower bound

C Lean listings

`lean/Bucket/Backtest.lean`, checked by `make lean`, holds the two facts the method leans on: the corrected permutation p has a positive numerator no larger than its denominator, and the length bands cover every word count.

```
structure PermutationP where
  hits : Nat
  permutations : Nat
  bounded : hits <= permutations

def PermutationP.numerator (p : PermutationP) : Nat := p.hits + 1
def PermutationP.denominator (p : PermutationP) : Nat := p.permutations
   + 1

theorem PermutationP.numerator_pos (p : PermutationP) : 0 < p.numerator
   :=
  Nat.succ_pos p.hits

theorem PermutationP.numerator_le_denominator (p : PermutationP) :
    p.numerator <= p.denominator :=
  Nat.succ_le_succ p.bounded

def lengthBand (words : Nat) : LengthBand :=
```

```
  if words < 20 then .short else if words <= 40 then .middle else .long

theorem lengthBand_total (words : Nat) :
    lengthBand words = .short \/ lengthBand words = .middle \/
       lengthBand words = .long
```

`lake build` output, quoted verbatim [empirical: papers/solvability-frontier/lean, 2026-10-08, make lean].

```
Build completed successfully (4 jobs).
```